COMSATS University Islamabad (CUI)

GitHub Repository Viva & Project Technical Report

Repository: https://github.com/furqanfayyaz/FA24-BCS-125

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Courses: Mobile Application Development & Web Technologies

Date: October 2026

1. Repository Overview

This repository contains all lab assignments, quizzes, and projects developed as part of the Mobile Application Development and Web Technologies coursework. Each module demonstrates proficiency in Flutter cross-platform mobile development (Stateful/Stateless widgets, local storage, CRUD operations, Material 3) and modern Web Technologies (HTML5 semantic markup, CSS3 external stylesheets, responsive tables, and forms).

2. Detailed Project Breakdown & Viva Explanations

Assignment 1: Salary Calculator App

A Flutter cross-platform mobile application designed to calculate net salary based on basic pay, allowances, tax deductions, and bonuses.

TextField, TextEditingController, StatefulWidget, Column, Row, ElevatedButton, Material 3 theme, number formatting.

• How do you capture user input? Using TextEditingController attached to TextField widgets.

• How is state managed? Using setState() inside a StatefulWidget to update calculations dynamically when the calculate button is pressed.

Assignment 2: Digital Student ID Card App

A professional digital student identity card application displaying student credentials in a secure card format with front and back side flipping, profile photo asset, and barcode simulation.

Card, ClipRRect, CircleAvatar, AssetImage, LinearGradient, Stack, Wrap, Expanded.

• How do you include local images in Flutter? Images are placed in the assets/ folder and declared under the flutter: assets section in pubspec.yaml.

• How does card flipping work? A boolean flag (_showBackSide) toggles between building the front widget tree and the back widget tree via setState().

Quiz 1: Student Registration Form

A Flutter screen demonstrating 10 core Flutter UI widgets arranged vertically in a Column with Center alignment, providing input controls for student registration.

Text, TextField, Radio, Checkbox, Option List (RadioListTile), Selection List (CheckboxListTile), DropdownButton, Simple Button, Submit Button.

• Why wrap a Column in SingleChildScrollView? To prevent layout overflow errors when the software keyboard appears on smaller mobile screens.

• What is the difference between Radio and Checkbox? Radio allows selecting only one option from a mutually exclusive group, whereas Checkbox allows multiple independent selections.

Quiz 2: Doctor Patient Management App (CRUD & Local DB)

A fully functional patient records management system for doctors supporting complete CRUD (Create, Read, Update, Delete) operations with local file persistence.

path_provider, Dart dart:io JSON serialization, ListView.builder, search and filter logic, Patient and PatientDocument models.

• How does local data persistence work without SQLite native dependencies? We use path_provider to get the application documents directory and serialize patient lists to local JSON files.

• How are doctor notes and medical documents linked? Each Patient object contains a list of PatientDocument items representing lab reports and prescriptions.

Lab Assignment 1: CUI Library Portal Website

A multipage responsive university library portal website built using pure HTML5, CSS3, semantic tags, tables, and forms without JavaScript.

Semantic elements (header, nav, main, section, article, aside, footer), CSS box-sizing: border-box, tables with colspan/rowspan, HTML5 form validation attributes.

• Why use semantic tags instead of generic tags? Semantic tags improve code readability, accessibility, and SEO by clearly describing the meaning of content sections.

• What does box-sizing: border-box do? It ensures padding and borders are included within the element's total width and height, preventing layout breakage across viewports.

3. Conclusion & Readiness

All projects in this repository have been rigorously tested with 0 lint errors and full test coverage. The code is modular, well-documented, clean, and fully prepared for submission and viva presentation.